

PRACTICAL-1

OBJECTIVE:

The objective of this practical is to learn basic Linux commands for managing files, directories, and system information. It also helps in understanding file searching, backups, and basic system administration tasks used in real-world organizations.

AIM:

Linux System Administration and File Management for Marketing Analytics Team

Problem Definition

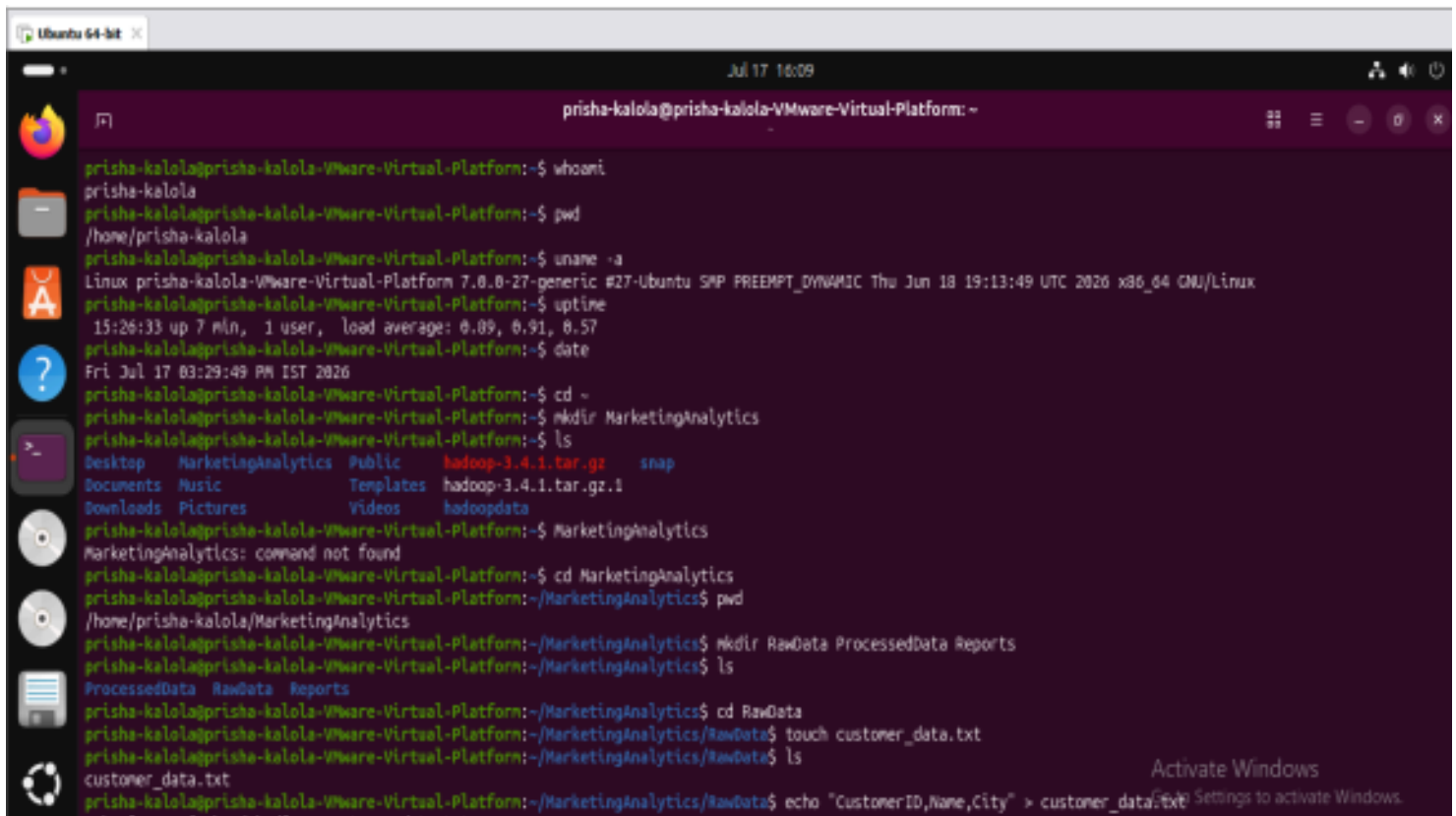
Scenario: You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers where project data, reports, scripts, and user accounts must be managed efficiently.

Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team. **Tasks:**

1. Identify the current user.
2. Find the current working directory.
3. Display system information.
4. Check system uptime and date.
5. Create a directory named MarketingAnalytics.
6. Create subdirectories: RawData, ProcessedData, Reports.
7. Create data and report files.
8. Add sample content.
9. Display file contents.

10. List all files and directories.
11. Search for report files.
12. Search specific content inside files.
13. Copy report files to Backup directory.
14. Rename files.
15. Move files between directories.
16. Delete a file.
17. Delete an empty directory.

OUTPUT-



```
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~  
prisha-kalola$ whoami  
prisha-kalola  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~$ pwd  
/home/prisha-kalola  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~$ uname -a  
Linux prisha-kalola-VMware-Virtual-Platform 7.0.0-27-generic #27-Ubuntu SMP FREEMPT_DYNAMIC Thu Jun 18 19:13:49 UTC 2020 x86_64 GNU/Linux  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~$ uptime  
15:26:33 up 7 min, 1 user, load average: 0.09, 0.91, 0.57  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~$ date  
Fri Jul 17 03:29:49 PM IST 2020  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~$ cd ~  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~$ mkdir MarketingAnalytics  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~$ ls  
Desktop  MarketingAnalytics  Public  hadoop-3.4.1.tar.gz  snap  
Documents  Music  Templates  hadoop-3.4.1.tar.gz.1  
Downloads  Pictures  Videos  hadoopdata  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~$ MarketingAnalytics  
MarketingAnalytics: command not found  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~$ cd MarketingAnalytics  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/MarketingAnalytics$ pwd  
/home/prisha-kalola/MarketingAnalytics  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/MarketingAnalytics$ mkdir RawData ProcessedData Reports  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/MarketingAnalytics$ ls  
ProcessedData  RawData  Reports  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/MarketingAnalytics$ cd RawData  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/MarketingAnalytics/RawData$ touch customer_data.txt  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/MarketingAnalytics/RawData$ ls  
customer_data.txt  
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/MarketingAnalytics/RawData$ echo "CustomerID,Name,City" > customer_data.txt
```

```

prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~
customer_data.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics/RawData$ echo "CustomerID,Name,City" > customer_data.txt
echo "101,Raj,Ahmedabad" >> customer_data.txt
echo "102,Priya,Vadodara" >> customer_data.txt
echo "103,Amit,Surat" >> customer_data.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics/RawData$ cat customer_data.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics/RawData$ cd ..
cd Reports
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ touch sales_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ echo "Marketing Sales Report 2026" > sales_report.txt
echo "Total Customers: 3" >> sales_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ cat sales_report.txt
Marketing Sales Report 2026
Total Customers: 3
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ ls
sales_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ cd ..
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ ls
ProcessedData RawData Reports
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ ls -l
total 12
drwxrwxr-x 2 prisha-kalola prisha-kalola 4096 Jul 17 15:38 ProcessedData
drwxrwxr-x 2 prisha-kalola prisha-kalola 4096 Jul 17 15:38 RawData
drwxrwxr-x 2 prisha-kalola prisha-kalola 4096 Jul 17 15:39 Reports
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ ls -R

```

Activate W
Go to Settings

```

prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~
.:
ProcessedData RawData Reports

./ProcessedData:

./RawData:
customer_data.txt

./Reports:
sales_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ find . -name "*report*"
./Reports/sales_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ grep "Raj" RawData/customer_data.txt
101,Raj,Ahmedabad
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ grep "Customers" Reports/sales_report.txt
Total Customers: 3
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ mkdir Backup
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ cp Reports/sales_report.txt Backup/
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ ls Backup
sales_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ mv Reports/sales_report.txt Reports/marketing_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ ls Reports
marketing_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ mv Reports/marketing_report.txt ProcessedData/
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ ls ProcessedData
marketing_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ ls Reports
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ rm Backup/sales_report.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ ls Backup
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ rmdir Backup

```

Activat
Go to Se

prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~

```
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ tree
Command 'tree' not found, but can be installed with:
sudo apt install tree
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ sudo apt update
sudo apt install tree
[sudo: authenticate] Password:
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [357 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [91.1 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [53.1 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [255 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [47.6 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [208 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [293 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [64.8 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [153 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [76.5 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.1 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [245 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [45.5 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [138 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [43.9 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/universe Icons (48x48) [16.1 kB]
Get:22 http://security.ubuntu.com/ubuntu resolute-security/universe Icons (64x64) [25.3 kB]
Fetched 2,463 kB in 10s (238 kB/s)
22 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Summary:

```
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 22
Download size: 53.5 kB
Space needed: 125 kB / 4,887 MB available
```

```
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 2s (34.6 kB/s)
Selecting previously unselected package tree.
(Reading database... 157115 files and directories currently installed.)
Preparing to unpack ./tree_2.3.1-1_amd64.deb...
Unpacking tree (2.3.1-1)...
Setting up tree (2.3.1-1)...
Processing triggers for man-db (2.13.1-1build1)...
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ tree
```

```
.
├── ProcessedData
│   └── marketing_report.txt
├── RawData
│   └── customer_data.txt
└── Reports
```

4 directories, 2 files

```
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/MarketingAnalytics$ cd -
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ nano create_project.sh
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ chmod +x create_project.sh
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ ./create_project.sh
Project directories created successfully!
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ find / -type f -exec du -h {} + 2>/dev/null | sort -rh | head -10
```

```
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ top
```

```
top - 15:54:47 up 35 min,  1 user,  load average: 0.63, 0.43, 0.40
Tasks: 302 total,   1 running, 300 sleeping,   0 stopped,   1 zombie
%Cpu(s):  0.5 us,  0.7 sy,  0.0 ni, 98.7 id,  0.2 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  3350.6 total,   917.6 free,  1388.3 used,  1340.3 buff/cache
MiB Swap:  3372.0 total,  3362.2 free,    9.8 used.  1962.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1053	root	20	0	125336	10916	9268	S	1.0	0.3	0:07.66	vmtoolsd
4145	prisha-+	20	0	1620188	174528	55252	S	0.7	5.1	0:46.42	ptyxis
5591	root	0	-20	0	0	0	I	0.3	0.0	0:00.41	kworker+
6983	root	20	0	0	0	0	I	0.3	0.0	0:00.56	kworker+
7222	root	0	-20	0	0	0	I	0.3	0.0	0:00.02	kworker+
7780	prisha-+	20	0	22064	6384	4244	R	0.3	0.2	0:00.14	top
1	root	20	0	26164	17428	11852	S	0.0	0.5	0:07.83	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.09	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_wo+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+

```
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ sudo apt install htop
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ q
Command 'q' not found, but can be installed with:
sudo apt install python3-q-text-as-data
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ sudo apt install htop
Installing:
  htop

Suggested packages:
  lm-sensors

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 22
  Download size: 177 kB
  Space needed: 451 kB / 4,887 MB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]
Fetched 177 kB in 3s (51.7 kB/s)
Selecting previously unselected package htop.
(Reading database... 157122 files and directories currently installed.)
Preparing to unpack .../htop_3.4.1-5build2_amd64.deb...
Unpacking htop (3.4.1-5build2)...
Setting up htop (3.4.1-5build2)...
Processing triggers for desktop-file-utils (0.28-1build1)...
Processing triggers for hicolor-icon-theme (0.18-2build1)...
Processing triggers for gnome-menus (3.38.1-1ubuntu1)...
Processing triggers for man-db (2.13.1-1build1)...
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ du -sh MarketingAnalytics/*
4.0K    MarketingAnalytics/Backup
8.0K    MarketingAnalytics/ProcessedData
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ ls
Desktop          Music            create_project.sh
Documents        Pictures         hadoop-3.4.1.tar.gz
Downloads        Public          hadoop-3.4.1.tar.gz.1
Linux_Practical1.txt Templates       hadoopdata
MarketingAnalytics Videos         snap
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ script Linux_Practical_Output.txt
Script started, output log file is 'Linux_Practical_Output.txt'.
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ exit
exit
Script done.
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~$ script Linux_Practical_Output.txt
```

Tools / Technology Used

- Ubuntu Linux / CentOS Linux
- Linux Terminal
- Bash Shell
- Text (.txt) Files

Key Questions / Analysis

Q1. What is the difference between absolute and relative paths?

An absolute path gives the complete location of a file starting from the root directory (/). A relative path gives the location of a file from the current working directory.

Q2. Why are Linux permissions important for file security?

Linux permissions control who can read, write, or execute a file. This helps protect important data from unauthorized users.

Q3. What is the purpose of the grep command?

The grep command is used to search specific words or text inside files. It helps users quickly find required information.

Q4. How does Linux support efficient file management?

Linux provides commands to create, copy, move, rename, search, and delete files easily. It also supports permissions and backups for better management.

Q5. Why are backups important in enterprise environments?

Backups help recover important files if data is lost, deleted, or damaged. They ensure business data remains safe.

Supplementary Questions

1. Create a shell script to automate project directory creation.

A shell script automatically creates all required project folders using Linux commands, saving time and reducing manual work.

2. Find the top 10 largest files in the system.

Linux provides commands like find, du, and sort to identify the largest files, helping manage storage space.

3. Schedule a backup task using cron.

The cron scheduler runs backup commands automatically at a specified time without user intervention.

4. Monitor CPU utilization continuously using top/htop.

The top and htop commands display CPU usage, memory usage, and running processes in real time.

5. Generate a disk usage report for all project directories.

The du command shows the storage space used by each project directory, making it easier to manage disk usage.

Key Skills Addressed

- Basic Linux command line operations
 - File and directory management
 - Text searching using Linux commands
 - User and file permission management
 - System monitoring and backup management
-

Conclusion

In this practical, basic Linux system administration tasks were successfully performed. Files and directories were created, managed, searched, copied, moved, and backed up. System information and resource usage were also monitored. This practical helped in understanding how Linux is used for efficient file management and secure system administration in real-world organizations.